Course Title	Computer Networks	L	T	P	C	Periods
Course Code	**UCSH-502**	**4**	**0**	**0**	**4**	**56**
Course Category	**DSC**					
Course Objective	1. To develop an understanding of modern network architectures from design and performance perspective. To introduce the student to the major concepts involved in wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs). 2. To provide an opportunity to do network programming To provide WLAN measurement ideas.					

Course Outcomes: Develop the skill set to

1. Explain the functions of the different layers of the OSI Protocol.
2. Draw the functional block diagram of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) to describe the function of each block.
3. For a given requirement (small scale) of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) design it based on the market available component
4. For a given problem related TCP/IP protocol developed the network programming.
5. Configure DNS DDNS, TELNET, EMAIL, File Transfer Protocol (FTP), WWW, HTTP, SNMP, Bluetooth, Firewalls using open source available software and tools.

Prerequisites for the Course: UCSH-401

Unit	Topic	Contents	Periods
1	Introduction	Introduction to computer networks, evolution of computer networks and its uses, Advantages and Disadvantages of Computer Network, reference models: OSI reference Models, TCP/IP Protocol Suite Networking fundamentals: Internet, Circuit switching vs Packet switching, ISPs, Delay and Loss in Packet Switched Networks	10
2	Local Area Network	LAN Architecture, LAN topologies- Bus/ Tree LAN, Ring LAN, Star LAN, Wireless LAN, Ethernet and Fast Ethernet, Token Ring	8
3	Application layer and data link layer	Application Layer Protocols: HTTP, FTP, SMTP, DNS Data link layer design issues, Flow Control- Stop and Wait, Error Detection, Error Control, error detection and correction, data link layer protocols,	10

		sliding window protocols, example of data link protocol- HDLC	
4	Medium access layer	Channel allocation problem, multiple access protocols, Introduction to ALOHA, CSMA/CD, CSMA/CA	8
5	The network layer	Introduction, Routers, Network layer concepts, shortest path routing, flooding, distance vector routing, link state routing (without algorithms), congestion control and quality of service, internetworking, IP, Ipv4 Addressing vs Ipv6	10
6	The transport layer	The transport layer services, elements of transport protocols, TCP and UDP, Brief introduction to presentation and session layer, E-mail	10
		Total	56

Reference: 1. Data Communication & networking: Forouzan, B. A. , Vth Ed, The McGraw-Hill Companies, Inc - 2017.
2. Data And Computer Communications, by William Stallings, VII edition, Pearson Education, 2005.

Suggested Readings:
1. Andrew S. Tanenbaum, Computer Networks, Fourth Edition, Pearson Education, 2003
2. Jim Kurose, Keith Ross, A Top Down Approach Featuring the Internet , Third Edition, Pearson Education, 2004